Initial Project Planning Template

Date	28 June 2026
Team ID	
Project Name	A Comprehensive Measure of Well-Being (HDI Prediction System)
Maximum Marks	5 Marks

Product Backlog, Sprint Schedule, and Estimation (4 Marks)

Use the below template to create a product backlog and sprint schedule

Sprint	Functional Requirement (Epic)	User Story Number	User Story / Task	Story Points	Priority	Team Meml
Sprint-1	**Entity-Relationship Diagram**	USN-1	As a team lead, I can design an ER diagram showing all entities and relationships for the HDI prediction system.	2	High	Spandhana Mondam
Sprint-1	**Pre-requisites**	USN-2	As a team lead, I can document all the links and resources required for the project setup.	1	Medium	Bhagyalatha Kadali
Sprint-1	**Project Flow**	USN-3	As a team lead, I can create a project workflow diagram showing all 8 epics in sequence.	2	High	Spandhana Mondam
Sprint-1	**Epic 1: Environment Setup**	USN-4	As a developer, I can install all required Python packages (NumPy, Pandas, Matplotlib, Scikit-learn, Flask).	2	High	Spandhana Mondam
Sprint-1	**Epic 1: Environment Setup**	USN-5	As a developer, I can set up the project folder structure with Dataset, Flask, and Training directories.	1	High	Spandhana Mondam
Sprint-1	**Epic 2: Import Libraries**	USN-6	As a developer, I can import all required libraries (NumPy, Pandas, Matplotlib, Seaborn) into the notebook.	2	High	Spandhana Mondam
Sprint-2	**Epic 3: Dataset**	USN-7	As a developer, I can download the HDI dataset from Kaggle containing 195 countries.	1	High	Pechetti Lilly Lakshmi
Sprint-2	**Epic 3: Dataset**	USN-8	As a developer, I can explore and understand the dataset structure, features, and data types.	2	High	Pechetti Lilly Lakshmi
Sprint-2	**Epic 3: Dataset**	USN-9	As a developer, I can perform data visualization using heatmaps and strip plots to identify patterns.	3	Medium	Pechetti Lilly Lakshmi
Sprint-2	**Epic 4: Preprocessing**	USN-10	As a developer, I can select dependent and independent variables for model training.	2	High	Pechetti Lilly Lakshmi
Sprint-2	**Epic 4: Preprocessing**	USN-11	As a developer, I can check and handle null values in the dataset using mean imputation.	2	High	Pechetti Lilly Lakshmi
Sprint-2	**Epic 5: Train/Test Split**	USN-12	As a developer, I can split the dataset into training and testing sets using train_test_split.	1	High	Spandhana Mondam
Sprint-2	**Epic 6: Model Training**	USN-13	As a developer, I can train a Linear Regression model using the prepared training dataset.	3	High	Spandhana Mondam
Sprint-2	**Epic 6: Model Training**	USN-14	As a developer, I can generate predictions and evaluate the model using R-squared score.	2	High	Spandhana Mondam
Sprint-3	**Epic 7: Save Model**	USN-15	As a developer, I can save the trained model using Pickle serialization as HDI.pkl.	1	High	Bhagyalatha Kadali
Sprint-3	**Epic 8: Flask App**	USN-16	As a developer, I can build the Flask backend to load the model and handle prediction requests.	3	High	Bhagyalatha Kadali
Sprint-3	**Epic 8: Flask App**	USN-17	As a developer, I can create HTML templates (home, input form, result) for the web interface.	3	High	Bhagyalatha Kadali
Sprint-3	**Epic 8: Flask App**	USN-18	As a developer, I can run and test the complete Flask application for accurate predictions.	2	High	Bhagyalatha Kadali

Sprint-3	**Conclusion**	USN-19	As a team lead, I can document the project conclusion summarizing the well-being measurement approach.	1	Medium	Bhagyalatha Kadali
Sprint-3	**Dashboard Extension**	USN-20	As a user, I can view an HDI Score Comparison dashboard showing top and bottom performing countries.	3	High	Spandhana Mondam
Sprint-3	**Dashboard Extension**	USN-21	As a user, I can view a World HDI Map dashboard with country classification and category filters.	3	High	Spandhana Mondam
Sprint-3	**Dashboard Extension**	USN-22	As a user, I can view an HDI Trend Over Years dashboard showing regional development progress.	3	Medium	Pechetti lilly lakshmi